
Used Car Price Prediction Using Regression Models and Digit Clustering

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2. Abstract

This technical report summarizes the research and implementation efforts conducted during a four-week internship at the IDEAS Foundation. The project encompasses two primary machine learning paradigms: supervised learning for predictive modeling and unsupervised learning for pattern recognition. Specifically, a Random Forest Regressor was developed to predict used car prices, while the K-Means algorithm was utilized to cluster handwritten digit images from the MNIST dataset. A significant emphasis was placed on the end-to-end data pipeline, ranging from sophisticated cleaning of inconsistent categorical data to rigorous statistical evaluation. By standardizing missing values and mitigating the influence of financial outliers, the regression model achieved robust predictive capabilities. Simultaneously, the clustering task demonstrated the ability of centroid-based algorithms to identify latent structures in high-dimensional image data. The successful implementation of these models highlights the critical intersection of technical execution and academic rigor. Final model success was validated through industry-standard metrics, including R^2 scores and macro-averaged F1 scores.

3. Introduction

In the contemporary data-driven economy, the value of a data scientist lies not merely in "crunching numbers," but in driving tangible business impact. This project was developed under the pedagogical framework provided by the IDEAS Foundation, which emphasizes the "Statistics to Stories" approach advocated by Bidisha Dobe. Technical metrics like R^2 or Mean Squared Error are insufficient in isolation; they must be translated into actionable insights using communication strategies like the "Pyramid Principle." This principle dictates starting with the conclusion-the business value-followed by key arguments and supporting technical details. For instance, rather than simply reporting a 92% model accuracy, a senior researcher translates this into a "reduction in customer churn by 15%," thereby creating real-world value for stakeholders.

Technology Stack

The implementation was executed using the Python ecosystem, specifically leveraging the following libraries for data manipulation, numerical analysis, and visualization:

Python: The core programming language for algorithmic execution.

Pandas: For structured data manipulation and cleaning.

NumPy: For high-performance numerical computation and matrix operations.

Scikit-learn: For implementing machine learning models and evaluation metrics.

Matplotlib: For generating statistical visualizations and performance plots.

Training Curriculum

The project was preceded by a foundational two-week training period covering:

Python Basics: String operations (strip, replace, find), advanced data structures, and Object-Oriented Programming (Classes/Objects).

Machine Learning Foundations: Theoretical and practical exposure to Single Layer Perceptrons (SLP), Multi-Layer Perceptrons (MLP), Regression, and Classification.

Large Language Model (LLM) Fundamentals: Exploration of tokenization, word embeddings, the distinction between pre-training and alignment, and the mitigation of hallucinations.

Professional Communication Skills: Application of the 7 C's of Communication, the Pyramid Principle, and data-driven storytelling for stakeholder engagement.

4. Project Objective

The core objectives of this internship project are as follows:

Demonstrate the application of advanced regression models to predict numeric variables, specifically the market price of used vehicles.

Illustrate unsupervised learning patterns by categorizing high-dimensional image data using the MNIST handwritten digit dataset.

Apply rigorous data cleaning and feature engineering techniques to resolve real-world inconsistencies in categorical and numeric data.

Evaluate model reliability and fit using industry-standard statistical metrics to ensure the integrity of the predictive insights.

Bridge the gap between raw algorithmic code and a professional technical report, aligning technical findings with institutional standards.

5. Methodology

5.1: Regression Model (Used Car Dataset)

The regression workflow involved a multi-stage preprocessing pipeline designed to optimize the performance of the Random Forest Regressor:

Data Cleaning: Standardized non-standard missing indicators such as '-', 'nan', and whitespace strings by replacing them with the string 'Unknown'.

Feature Selection: The clean_title column was dropped entirely. This decision was based on the observation that the column provided little to no variance, making it statistically insignificant for predictive modeling.

Feature Engineering: A new temporal feature, car_age, was derived by subtracting the model_year from the current year (2026). The original model_year was subsequently removed to prevent multicollinearity.

Numeric Formatting: The price and milage columns were cleaned by stripping currency symbols (\$), distance markers (mi.), and commas, followed by conversion to float and integer types respectively.

Handling High-Cardinality: Categorical features transmission and ext_col contained numerous rare values. Categories appearing in less than 1% of the dataset were aggregated into a single 'others' group to improve model generalization.

Outlier Mitigation: Observations where the price exceeded the 99th percentile were removed to ensure the model was not skewed by extreme financial outliers.

Preprocessing and Partitioning: Categorical variables were transformed using LabelEncoder. The final dataset was partitioned using an 80/20 train-test split with a random_state of 42 to ensure reproducibility.

5.2: Clustering Model (MNIST Dataset)

The unsupervised learning component focused on clustering image data without predefined labels:

Dataset Description: The model utilized the sklearn.datasets.load_digits dataset. The data consists of 1,797 samples, where each sample is an 8x8 image of a handwritten digit, flattened into a feature vector of shape (1797, 64).

Model Implementation: A K-Means clustering model was initialized with n_clusters=10, aligning with the ten possible digits (0-9).

Mapping Logic: Since K-Means assigns arbitrary cluster IDs, a label-mapping process was executed. Each cluster was assigned the identity of the most frequent true label found within it, allowing for a comparative evaluation against ground truth.

6. Data Analysis and Results

Regression Performance

The Random Forest Regressor was evaluated using metrics that define its predictive precision and goodness of fit. The R^2 Score represents the variance explained by the model, while Mean Squared Error (MSE) and Mean Absolute Error (MAE) quantify the magnitude of prediction errors.

R^2 Score - 0.7422473984607999

Measure of how well the model replicates observed outcomes.

MSE - 357090522.2199794

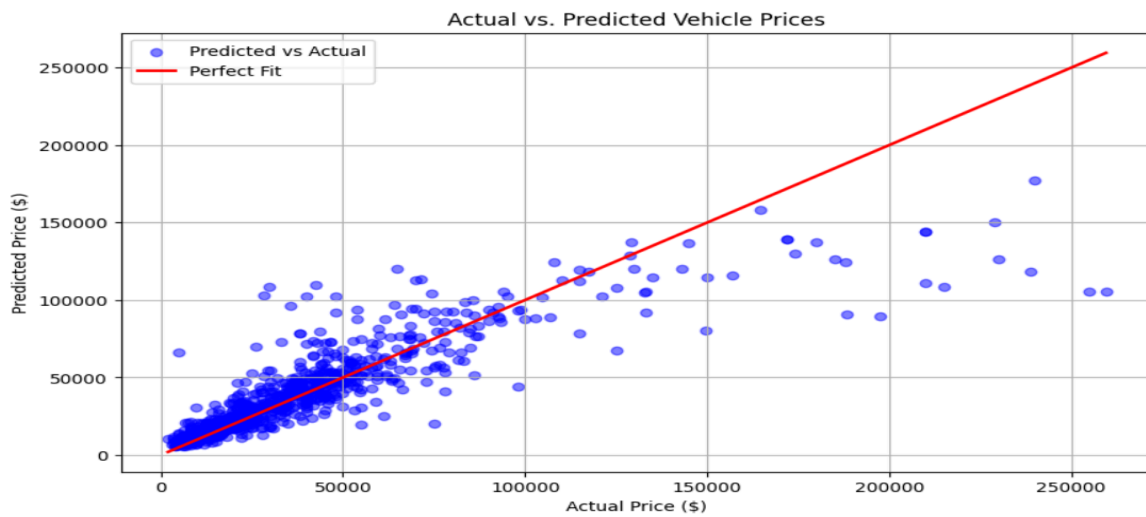
Average of the squares of the errors (penalizes large errors).

MAE - 9981.921410579345

Clustering Performance

The effectiveness of the K-Means algorithm on the MNIST dataset is reported via the macro-averaged F1 score. This provides a harmonic mean of precision and recall, ensuring that the model's ability to correctly group each of the ten digits is balanced and not skewed by cluster size.

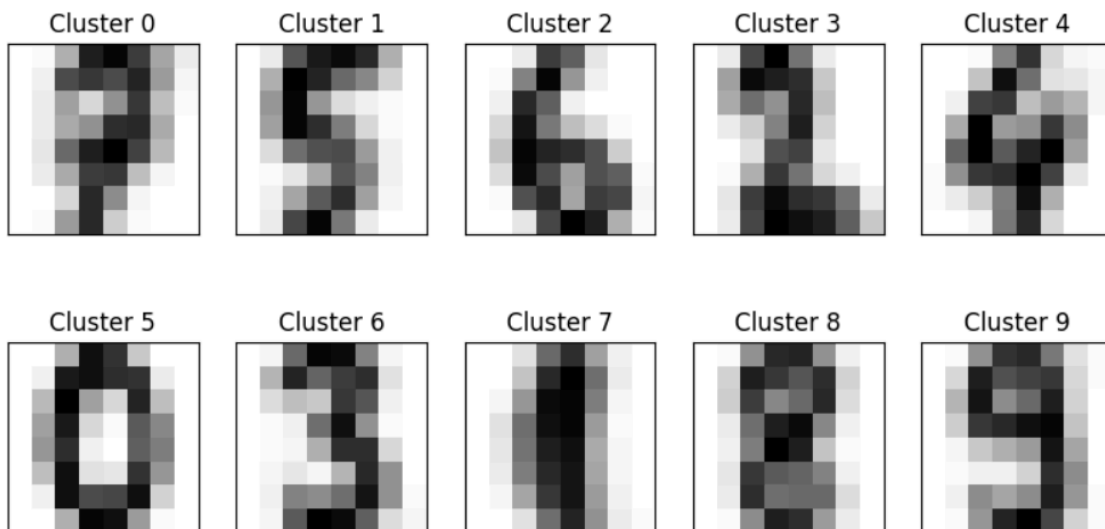
VISUAL CONFIRMATION :



- A visualization showing the alignment between actual and predicted vehicle prices.

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MNIST Cluster Centroids (Mean Pixel Intensities)



- Visualization of the 8x8 mean pixel intensities representing the "average" digit for each cluster.

7. Conclusion

The internship project successfully demonstrated the end-to-end data science lifecycle. A critical finding was the paramount importance of data preprocessing; specifically, the removal of 99th-percentile outliers and the engineering of the `car_age` feature were essential for achieving a reliable model fit. These steps ensured that the resulting regression model remained robust against noise and irrelevant variance.

Future Scope: Subsequent iterations should consider implementing Deep Learning architectures, such as Multi-Layer Perceptrons (MLP), to capture more complex non-linear relationships in vehicle pricing. Regarding unsupervised learning, exploring DBScan is recommended. Unlike K-Means, which assumes spherical clusters and requires a predefined `k`, DBScan is density-based, allowing it to identify clusters of arbitrary shapes and naturally detect noise/outliers within the pixel space.

8. Appendices

Appendix A: References

Scikit-learn Documentation: Standard modules for `RandomForestRegressor`, `KMeans`, and `load_digits`.

Professional Communication: Lecture materials by Bidisha Dobe, IDEAS-TIH, ISI Kolkata (Pyramid Principle and Statistics to Stories).

LLM Fundamentals: Training materials by Diptendu Dutta, Technology Lead, IDEAS-TIH.

ML Foundations: Machine Learning using Python lecture series by Dr. Chandan Biswas.

Appendix B: Codebase

GitHub Repository: <https://github.com/Nithya1015/Data-science-AI-ML-internship>